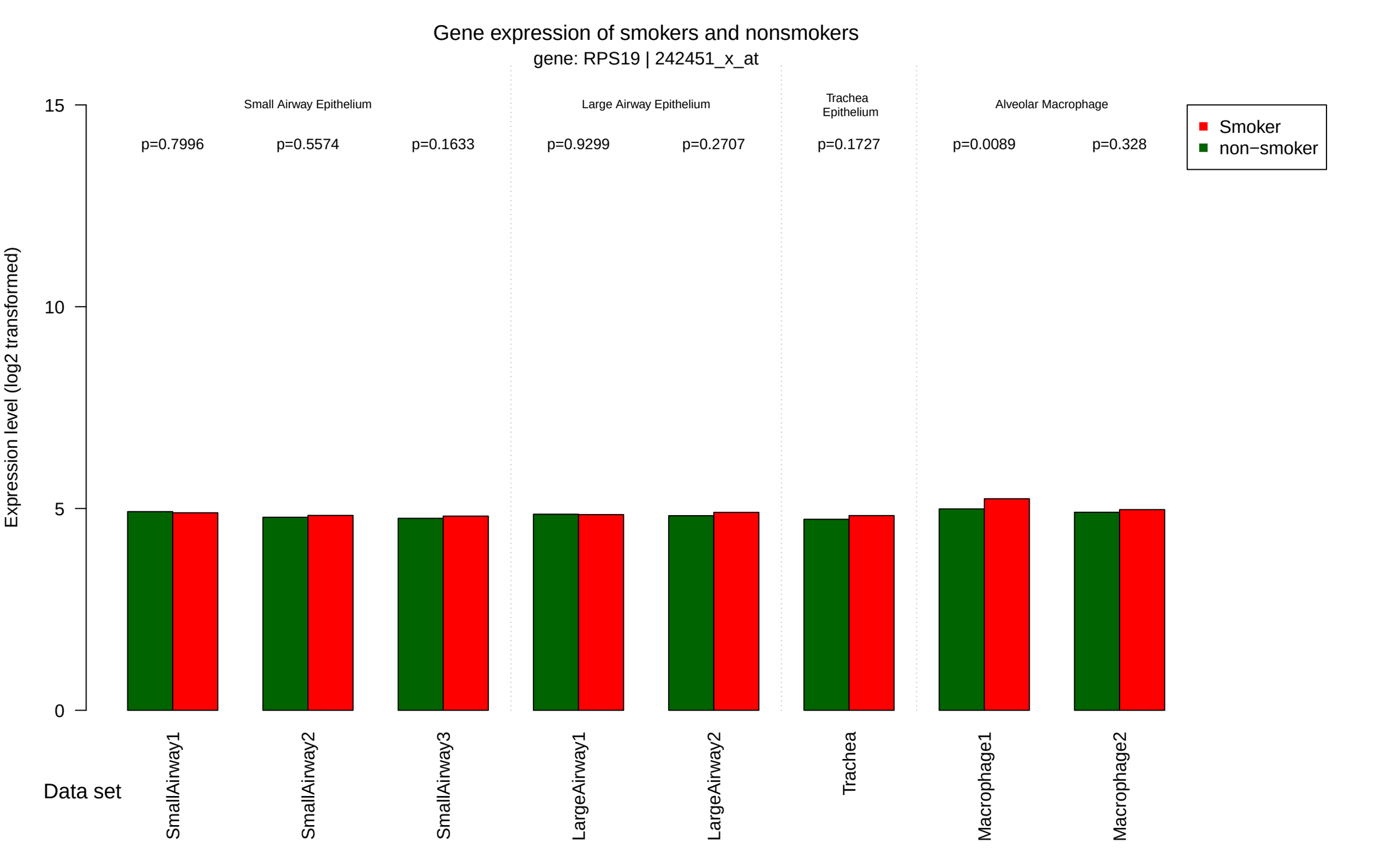
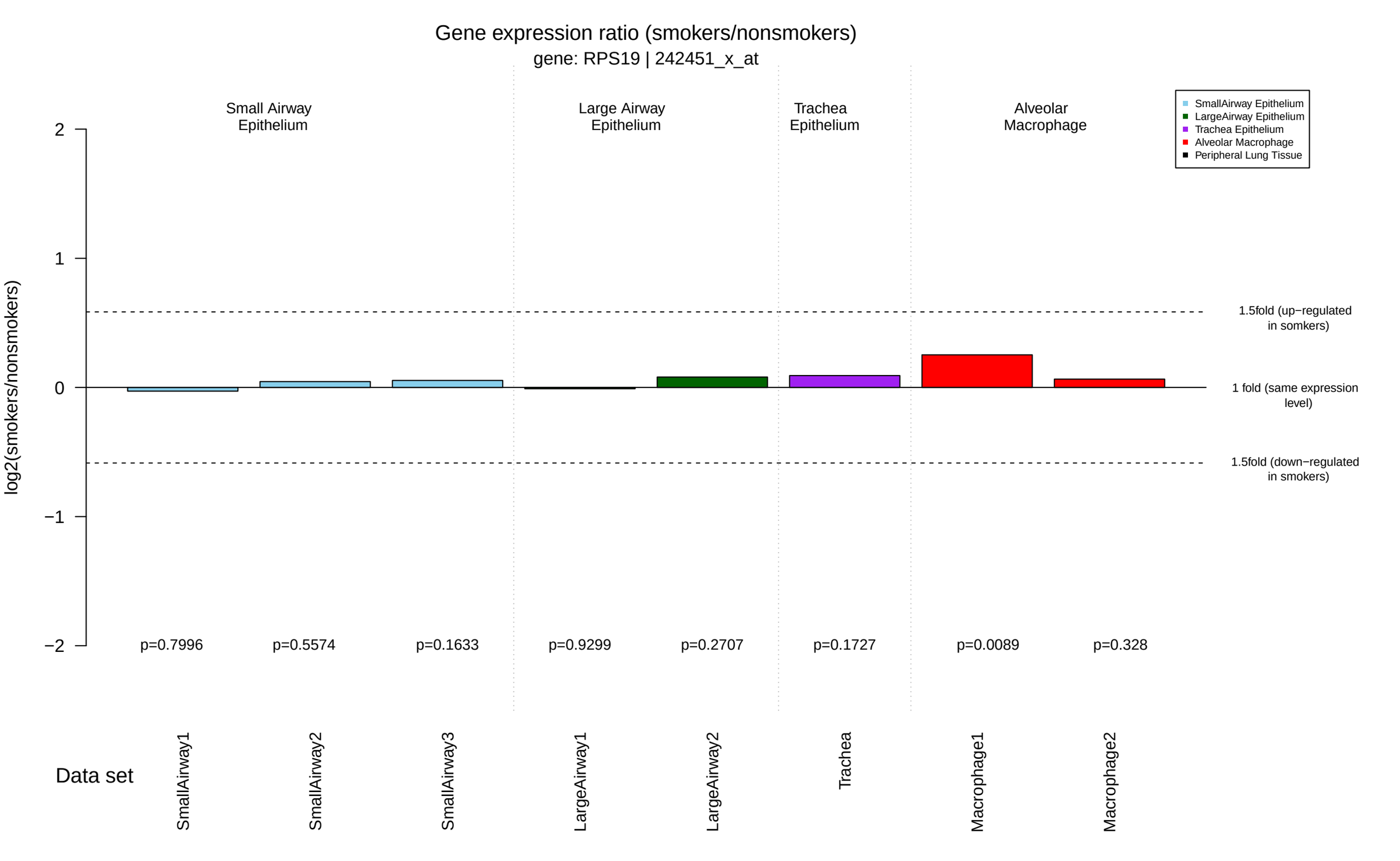
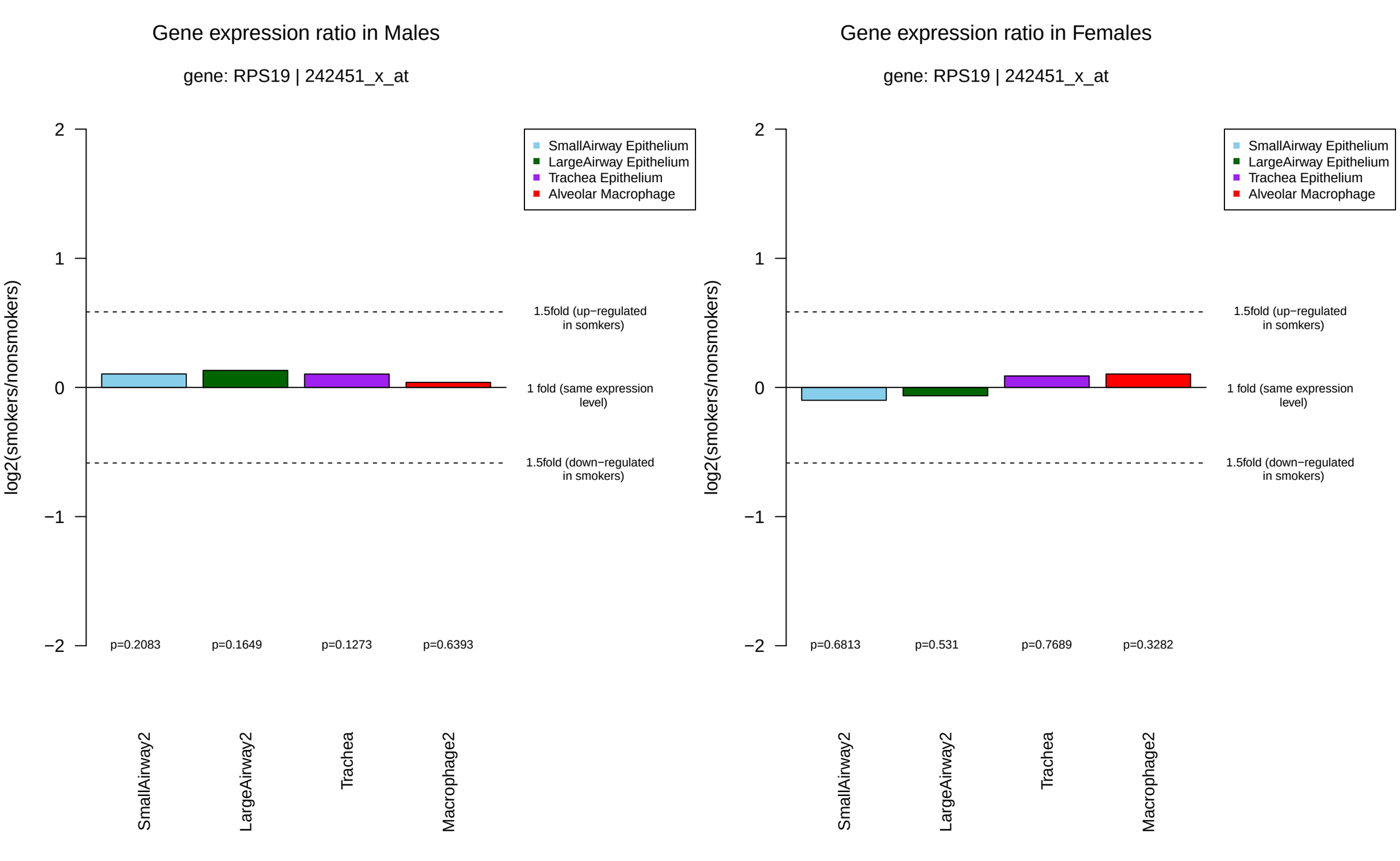
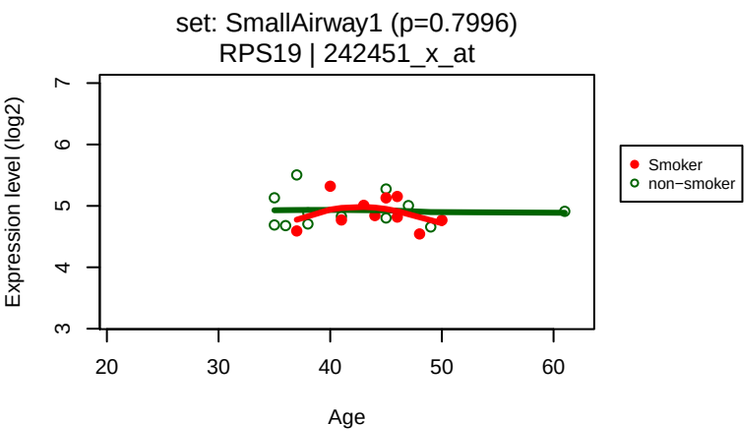
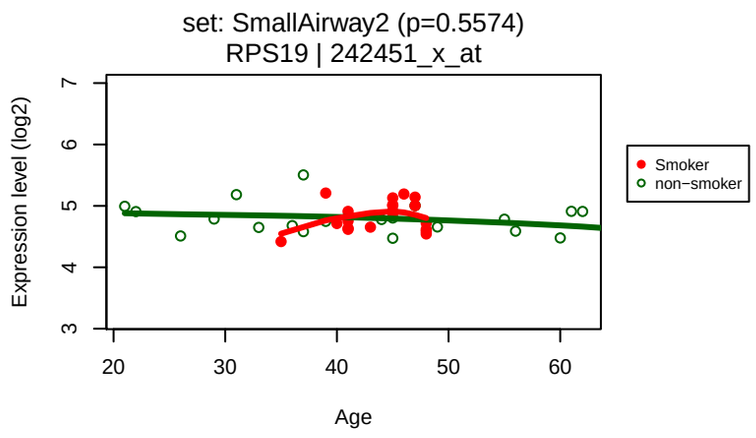
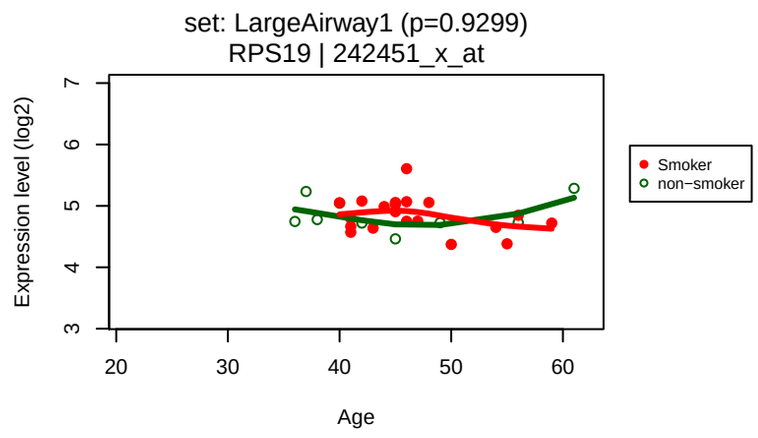

Gene expression of smokers and nonsmokers
gene: RPS19 | 242451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7996
p=0.5574
p=0.1633
p=0.9299
p=0.2707
p=0.1727
p=0.0089
p=0.328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

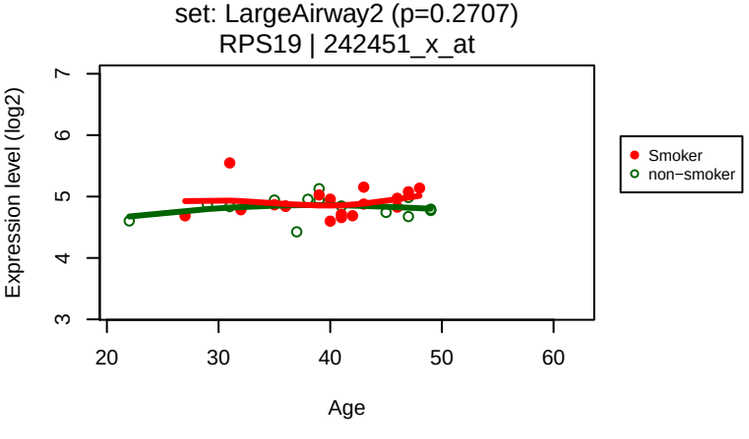

Gene expression ratio (smokers/nonsmokers)
gene: RPS19 | 242451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7996
p=0.5574
p=0.1633
p=0.9299
p=0.2707
p=0.1727
p=0.0089
p=0.328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

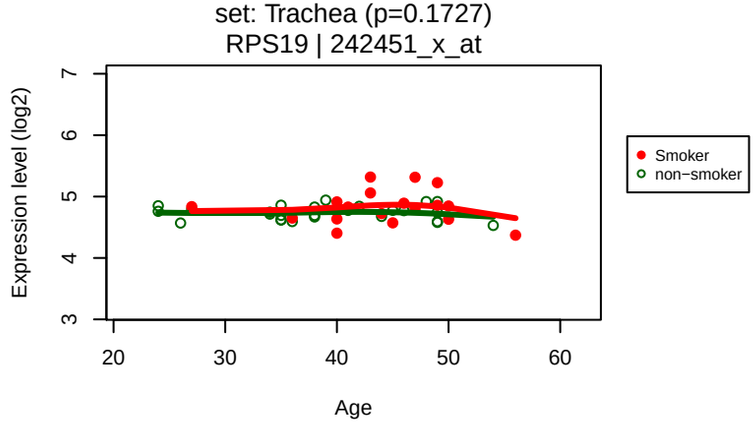

Gene expression ratio in Males
gene: RPS19 | 242451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2083
p=0.1649
p=0.1273
p=0.6393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS19 | 242451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6813
p=0.531
p=0.7689
p=0.3282
SmallAirway2
LargeAirway2
Trachea
Macrophage2

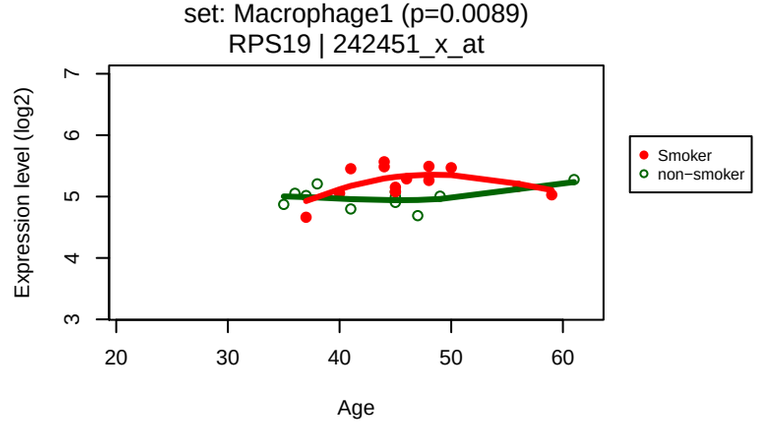

set: SmallAirway1 (p=0.7996)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker

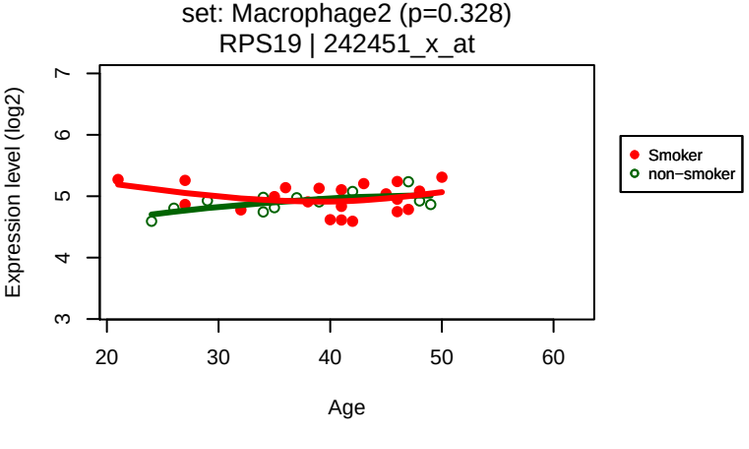

set: SmallAirway2 (p=0.5574)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker

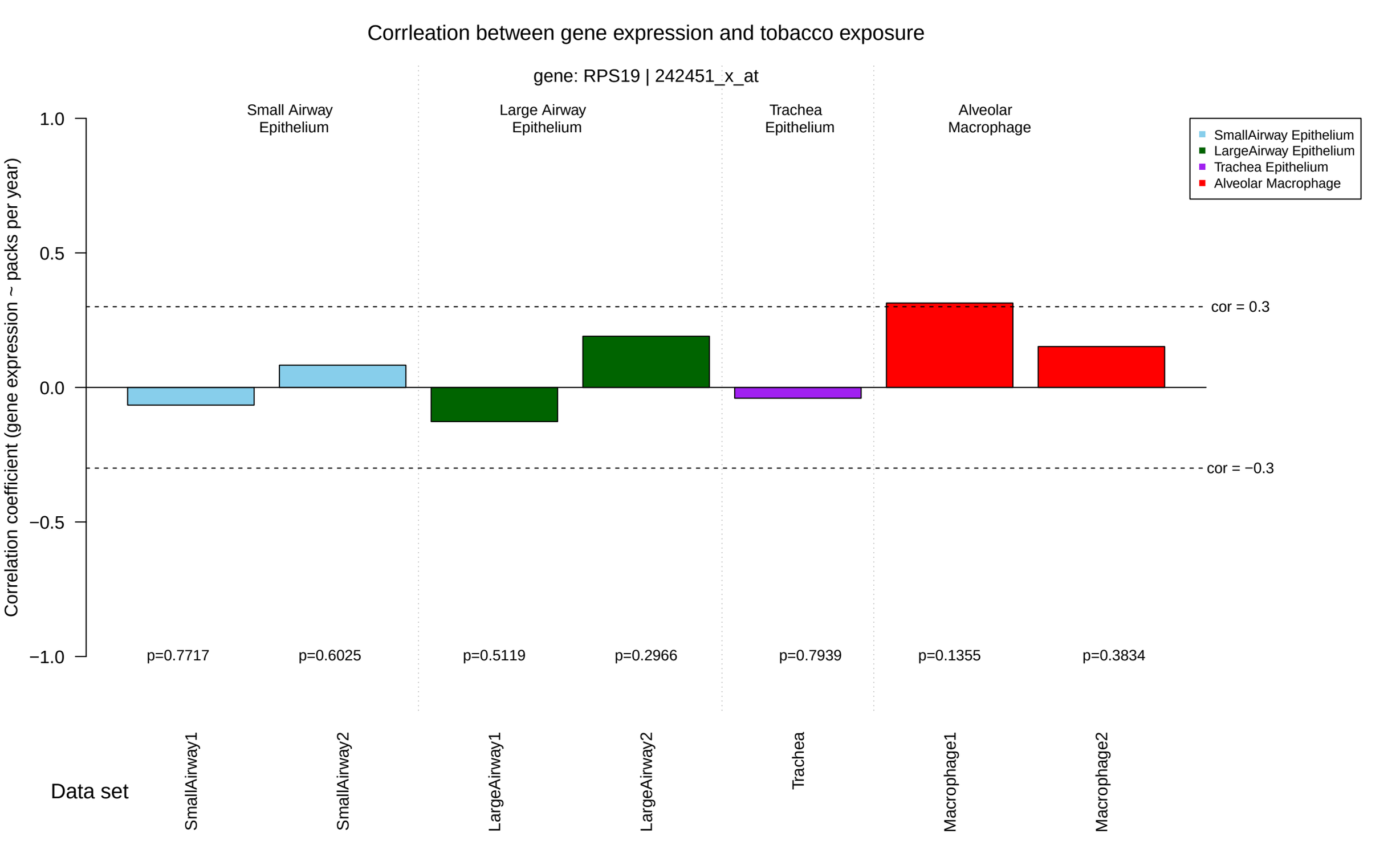

set: LargeAirway1 (p=0.9299)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2707)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1727)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0089)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.328)
RPS19 | 242451_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS19 | 242451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7717
p=0.6025
p=0.5119
p=0.2966
p=0.7939
p=0.1355
p=0.3834
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2